STCFCM: A Spatial and Temporal Cloud Fraction-Based Calibration Method for Satellite-Derived Near-Infrared Water Vapor Product

Jiafei Xu and Zhizhao Liu, *Member, IEEE*

***Abstract*— Precipitable water vapor (PWV) data from satellite-sensed near-infrared (NIR) measurements offer a unique source for monitoring atmospheric water vapor distribution locally and globally. However, the observational quality of satellite-based operational NIR PWV products is considerably affected by the presence of clouds. We develop a spatial and temporal cloud fraction-based calibration method (STCFCM) to calibrate satellite-sensed NIR PWV products and improve PWV accuracy. The STCFCM is built based on a light gradient boosting machine (LightGBM) using cloud fraction, together with spatial–temporal fields—latitude, longitude, height, and month. The newly calibrated PWV estimates from the MODIS sensor onboard the Terra satellite show that the STCFCM-estimated PWV data exhibit a better agreement with reference PWV estimates from global navigation satellite system (GNSS) and radiosonde observations. The root-mean-square error (RMSE) of MODIS/Terra operational PWV products is reduced by 55.53% from 11.40 to 5.07 mm compared to GNSS PWV and 60.74% from 14.67 to 5.76 mm compared to radiosonde PWV. The calibrated all-weather PWV estimates outperform operational clear-sky PWV products, highlighting the effectiveness of the STCFCM in calibrating satellite-sensed NIR PWV retrievals, particularly for cloudy sky conditions. The newly developed STCFCM approach is the first one in the research community to calibrate global PWV data from satellite NIR measurements. It is a promising technique to calibrate remote-sensing PWV products from other satellite observations under all weather conditions.**

***Index Terms*— Calibration, cloud fraction, global navigation satellite system (GNSS), near-infrared (NIR), precipitable water vapor (PWV), radiosonde.**

I. Introduction

WATER vapor is a crucial atmospheric greenhouse parameter [1], which has an indispensable impact on the Earth's weather [2], climate [3], and environment [4], [5], [6] on regional and global scales. The distribution of atmospheric water vapor presents a considerable variation in both spatial and temporal dimensions [7], [8]. Water vapor is also a key factor in numerical weather prediction (NWP) modeling [9], [10]. NWP is a widely used approach to utilize current observations to generate future forecasts of various meteorological elements, such as wind, temperature, precipitation, and so forth [11], [12].

Several water vapor-monitoring instruments have been developed to measure atmospheric precipitable water vapor (PWV) estimates, including sun-photometer, Lidar, radiosonde, and global navigation satellite system (GNSS) [13], [14], [15]. Among them, the ground-based GNSS instrument can observe high temporal-resolution atmospheric PWV data with GNSS stations across the globe [16], literally independent of weather conditions [17]. GNSS-based water vapor measurements agree well with PWV estimates from radiosonde measurements [18], [19], and they are used as a reliable water vapor reference to test other PWV measurements [20], [21] and considered as the desired resultant water vapor estimates in algorithm development procedure [22], [23], [24]. However, GNSS instruments can only provide station-wise PWV data records worldwide.

Different from ground-based instruments, satellite remote-sensing sensors offer an efficient method to observe the local or global distribution of water vapor at spatial–temporal dimensions. Satellite-measured PWV can be retrieved at different spectral wavelength channels [25], [26], [27], [28]. Among them, satellite-sensed PWV data measurements from near-infrared (NIR) channels have been the most common water vapor measurement technique, since NIR bands have a large PWV sensitivity at the atmospheric boundary layer, where most PWV exists [29]. Yet, clouds have a significant impact on the measurement performance of satellite-based NIR PWV data records, as satellite NIR channels cannot go through the clouds [16], [30].

The Moderate Resolution Imaging Spectroradiometer (MODIS), deployed on the Aqua and Terra satellites, can offer PWV measurements over both land and ocean based on NIR channels [29]. The MODIS NIR cloudy-sky water vapor estimates presented a root-mean-square error (RMSE) of above 14 mm compared to GNSS-derived water vapor measurements, which is much larger than MODIS-derived NIR confident-clear water vapor observations (RMSE = 4–5 mm) [20]. Under both clear and cloudy sky conditions, that is, all-weather, the RMSE between MODIS NIR water

Manuscript received 12 September 2023; revised 5 January 2024 and 19 March 2024; accepted 23 March 2024. Date of publication 27 March 2024; date of current version 12 April 2024. This work was supported by the Research Grants Council of the Hong Kong Special Administrative Region, China, under Project PolyU/RGC 15221620/B-Q80Q, Project 15205821/B-Q84W, and Project 15212622/B-Q94L. *(Corresponding author: Zhizhao Liu.)*

The authors are with the Department of Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University, Hong Kong (e-mail: garfield.xu@connect.polyu.hk; lszzliu@polyu.edu.hk).

Digital Object Identifier 10.1109/TGRS.2024.3382312

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:33:45 UTC from IEEE Xplore. Restrictions apply.

vapor and GNSS water vapor is around 10 mm, because of the impact of clouds [20]. In addition to clouds, the observational performance of satellite-based NIR water vapor products depends significantly upon location, solar zenith angle, season, dry-wet condition, and water vapor values [20]. It should be mentioned that the cloud and PWV factors have a larger influence on satellite NIR PWV observations, compared to other dependence parameters [30].

Many efforts have been conducted to calibrate the observational quality of operational PWV products from satellite-sensed NIR measurements. The recent study by Xu and Liu [31] proposed a machine-learning-based calibration model to calibrate the Sentinel-3 operational NIR water vapor estimates, which showed a decrease in RMSE of more than 20% in China and its surrounding areas. Zhu et al. [24] proposed a grid-based calibration approach to calibrate the performance of MODIS-based NIR clear-sky water vapor measurements. The results showed that the RMSE between PWV from MODIS and GNSS decreased to 0.60–4.30 mm across China. In our recent studies in Xu and Liu [32], [33], we developed several calibration methods based on machine learning and conventional regression to refine the observational performance of operational MODIS NIR water vapor data over Australia and China, with a decrease in RMSE of more than 50% with respect to GNSS-observed reference PWV. The previous research of calibrating satellite-based NIR water vapor products, in general, had a focus on regional scales, that is, China or Australia, though operational water vapor measurements from satellite NIR channels are applicable globally.

In this article, a new spatial–temporal cloud fraction-based calibration model (STCFCM) is developed to calibrate global PWV data from satellite NIR measurements for the first time. In addition, the STCFCM model has considered the impact of cloud factors. The STCFCM is established based on the light gradient boosting machine (LightGBM) as well as ground-based GNSS-derived PWV observations. The input of the STCFCM consists of NIR PWV from the MODIS sensor, cloud fraction, spatial information (latitude, longitude, and elevation), and temporal information (month), which are strongly correlated with the observational performance of satellite-retrieved NIR water vapor data. The STCFCM model can compute the calibrated MODIS NIR PWV estimates based on LightGBM considering cloud fraction, in addition to the spatial and temporal fields.

This article is organized as follows. Section II presents the GNSS, MODIS, and radiosonde datasets across the world, which are employed in the development and validation of the STCFCM. A detailed description of the STCFCM model is presented in Section III. In Section IV, the global validation of both operational and calibrated MODIS NIR water vapor estimates is conducted based on water vapor observations from independent 5465 GNSS stations in 2017–2019. In addition, the verification of the newly calibrated PWV data is also conducted using radiosonde-observed reference PWV during 2017–2019 in Section IV. The validation result is discussed in Section V, and Section VI summarizes the major findings of this research.

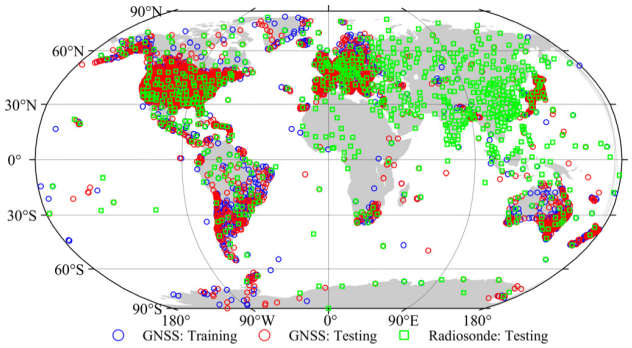


Fig. 1. Geographical distribution of 5000 GNSS stations (red circle) used for algorithm development and 5465 GNSS stations (blue circle) used for algorithm evaluation. A total of 783 radiosonde stations (green square) are also used to verify the performance of the STCFCM.

II. Data and Preprocessing

A. MODIS

PWV can be estimated from daytime NIR measurements of the MODIS sensor onboard the Aqua and Terra satellites [34]. The MODIS sensor observes the whole globe in 36 spectral channels from 0.4 to 14.4 μm in 1–2 days [34], [35]. The MODIS-observed NIR water vapor data are retrieved based on five NIR channels, that is, three PWV absorption bands as well as two window channels [29]. A ratio method based on 2-band or 3-band has been used to obtain atmospheric water vapor from MODIS NIR observations over land, ocean, and cloud regions with sun glint [36].

In this article, three types of MODIS data products from the Terra satellite are used: MOD05 water vapor, MOD03 geolocation, and MOD06 cloud datasets. The MOD03 spatial geolocation fields include, that is, latitude, longitude, and height. The MOD06 cloud product is utilized to obtain the cloud fraction information of MODIS water vapor observations. Four categories can be classified according to the probability of cloud fraction level: $<$1% (confident clear), $<$5% (probably clear), $<$34% (probably cloudy), and $\geq$34% (confident cloudy) [37].

B. GNSS

The worldwide PWV measurements, derived from GNSS stations by the Nevada Geodetic Laboratory (NGL) [38], are utilized for algorithm training and validation. Fig. 1 lists the geographical distribution of ground-based GNSS stations across the world. Among them, 5000 stations are utilized for the training of the STCFCM (see Fig. 1, blue circles), whereas the other 5465 stations are used for model validation (see Fig. 1, red circles).

The NGL-based GNSS data measurements are processed using the GipsyX package [38], [39]. The NGL GNSS tropospheric delay measurements are computed as a combination of dry zenith delay and wet zenith delay based on a Vienna Mapping Function 1 (VFM1) [38]. The NGL GNSS water vapor estimates are derived from the wet zenith delay based on the retrieval approach illustrated by Bevis et al. [40], [41].

In this research, the NGL global GNSS-derived PWV estimates, observed from 5000 stations in 2017, are employed as the expected resulting water vapor observations for STCFCM

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:33:45 UTC from IEEE Xplore. Restrictions apply.

model development. The other 5465 GNSS stations distributed globally in 2017–2019 are used to evaluate the performance of the STCFCM.

C. Radiosonde

In addition, PWV data estimated from 783 radiosonde stations during 2017–2019, collected from the Integrated Global Radiosonde Archive Version 2 (IGRA2) [42] and displayed in Fig. 1, are also utilized as a reference to evaluate the performance of the STCFCM model. One radiosonde station normally has PWV measurements at 0 and 12 UTC.

The PWV from the radiosonde can be calculated based on specific humidity observations from the surface to the top of the sounding. In this study, the IGRA2 radiosonde-based PWV estimates are retrieved using the algorithm proposed by Zhang et al. [43].

D. Matching Criteria

To perform the algorithm development and validation, two main criteria are employed to select spatial–temporal collocation pairwise PWV observations from MODIS and GNSS as well as MODIS and radiosonde.

In the spatial dimension, MODIS pixels are considered if the distance from the pixel center to the closest GNSS/radiosonde station does not surpass 5 km. The temporal difference between MODIS and GNSS measurements is not larger than 30 min. The temporal difference between paired MODIS–radiosonde data observations is required to be below 3 h. Both spatial and temporal collocation criteria listed above are based on previous studies [16], [28], [44].

III. Methods

A. Light Gradient Boosting Machine (LightGBM)

LightGBM is a distributed and efficient gradient boosting method, which utilizes decision tree-based learning approaches [45]. It is a new implementation of the gradient boosting decision tree (GBDT) method [45], [46]. Both LightGBM and GBDT use decision trees as weak learners, reducing the biases through the residual of prior decision trees. Both can be used to address regression problems [45], [46].

In contrast to other decision tree-based learning methods that grow trees based on depth-wise, the LightGBM grows trees using the best-first with the smaller loss [45]. The histogram-based approach of the LightGBM can help speed up training and reduce memory usage, compared with other boosting algorithms that use presort-based methods [45]. The final estimation result of the LightGBM is a weighted mean value from all output values of each decision tree [45].

B. Spatial–Temporal Cloud Fraction-Based Calibration Model (STCFCM)

To calibrate the operational water vapor estimates from satellite MODIS NIR bands, an STCFCM model is developed based on the global GNSS PWV data of 2017 in this article. The STCFCM is established using a machine-learning model, that is, LightGBM.

In this study, the differential water vapor data is estimated by subtracting GNSS-estimated PWV from satellite-based MODIS NIR water vapor. It is written as

$$\mathrm{PWV}_D = \mathrm{PWV}_S - \mathrm{PWV}_G \tag{1}$$

where PWV_D is the differential PWV; PWV_S is the operational water vapor from satellite MODIS NIR measurements, that is, MOD05 water vapor; and PWV_G is the reference water vapor from ground-based GNSS measurements.

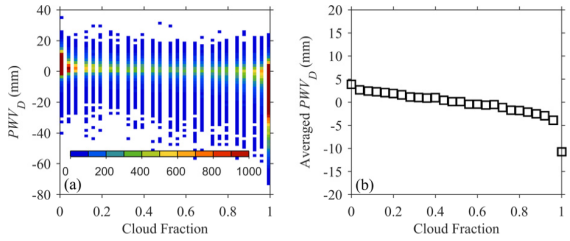


Fig. 2. Differential water vapor data between MODIS NIR PWV and GNSS PWV under different cloud fraction levels. The GNSS PWV data, observed from global 5000 training stations in 2017, are used to estimate the differential PWV. (a) Scatter plot of differential PWV versus cloud fraction. (b) Scatter plot of averaged differential PWV versus cloud fraction. The color bar indicates the number of data points.

The differential PWV, that is, PWV_D, is associated with many influence factors—PWV, location, cloud, and season, particularly the impact of clouds [16], [20]. Fig. 2 shows the differential PWV_D based on global 5000 training stations in 2017 (see Fig. 1), under different cloud fraction conditions. It can be observed in Fig. 2(a) that the PWV_D varies significantly under different cloud fraction levels as well as under the same cloud fraction level. By averaging PWV_D at each different cloud fraction level, the mean PWV_D showed a good relationship with the cloud fraction levels, as the result presented in Fig. 2(b).

The estimation of PWV_D is considered a regression problem to relate the differential PWV with the MODIS NIR PWV, cloud fraction, latitude, longitude, height, and month. It is a great challenge to address this complex multiparameter regression problem based on conventional methods. Instead, a machine-learning approach, that is, LightGBM, is used to describe the functional relationship between the resulting PWV_D and multiple input factors.

The GBDT presents a good performance in calibrating satellite-sensed all-weather PWV observations [33], [47]. Compared with other machine-learning approaches, the GBDT model also exhibits a better performance to improve the accuracy of satellite-derived PWV products [33]. The LightGBM, that is, a new implementation of the GBDT, has higher training speed, reduced memory usage, and smaller loss, compared to other boosting machine-learning models, such as GBDT [45], [46]. As a result, the LightGBM machine-learning approach is selected in this study to address the complex regression between PWV_D and multiple input elements.

The STCFCM-estimated PWV_D is calculated as

$$\mathrm{PWV}_D = \mathrm{LightGBM}\left(\mathrm{PWV}_S, C_f, L_t, L_g, H_t, M_n\right) \tag{2}$$

where C_f is the cloud fraction field of MODIS observations; L_t, L_g, and H_t are the spatial latitude, longitude, and height fields of MODIS observations, respectively; and M_n is the temporal month number of MODIS observations in a year.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:33:45 UTC from IEEE Xplore. Restrictions apply.

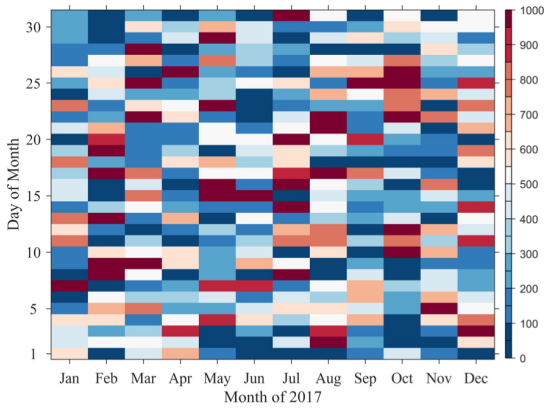


Fig. 3. Number of matched MODIS–GNSS data records in 2017 across the world, which are used for the training of the STCFCM. The color bar indicates the number of data points.

Once the differential PWV (i.e., PWV_D) is obtained, we can estimate the MODIS-calibrated NIR water vapor product (PWV_A) by subtracting the PWV_D from the PWV_S. It is defined as

$$\mathrm{PWV}_A = \mathrm{PWV}_S - \mathrm{PWV}_D. \tag{3}$$

In this research, the global PWV estimates, measured from 5000 GNSS stations in 2017, are used to train the STCFCM (see Fig. 1, blue circles). The number of collocated MODIS–GNSS data measurements used in model training is presented in Fig. 3. In the STCFCM training procedure, we select the GBDT as the boosting method in the LightGBM model. The optimization of LightGBM-based parameters is conducted based on a fivefold cross-evaluation approach.

Fig. 4 shows the flowchart for developing and validating the STCFCM model, which is built using the LightGBM machine-learning method. This STCFCM model can obtain MODIS-calibrated NIR water vapor estimates, without the consideration of cloud conditions.

C. Assessment Metrics

The performance of the STCFCM is validated utilizing worldwide GNSS PWV data from an additional 5465 stations from 2017 to 2019 as well as worldwide radiosonde PWV data from 783 stations from 2017 to 2019 (see Fig. 1). Three statistical metrics are used, that is, correlation coefficient (R^2), RMSE, and MB. They are calculated as

$$R^2 = \left[\frac{\sum_{i=1}^{N}\left(\mathrm{WV}_S - \overline{WV}_S\right)\left(\mathrm{WV}_G - \overline{WV}_G\right)}{\sqrt{\sum_{i=1}^{N}\left(\mathrm{WV}_S - \overline{WV}_S\right)^2\left(\mathrm{WV}_G - \overline{WV}_G\right)^2}}\right]^2 \tag{4}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\mathrm{WV}_S - \mathrm{WV}_G)^2} \tag{5}$$

$$\mathrm{MB} = \frac{1}{N}\sum_{i=1}^{N}(\mathrm{WV}_S - \mathrm{WV}_G) \tag{6}$$

where N is the number of collocated MODIS–GNSS or MODIS–radiosonde measurements utilized for the assessment of the STCFCM; WV_S is water vapor estimates from satellite MODIS observations; $\overline{WV}_S$ is the average of water vapor estimates from satellite MODIS observations; WV_G is reference water vapor estimates from GNSS/radiosonde measurements; and $\overline{WV}_G$ is the average of reference water vapor estimates from GNSS/radiosonde measurements.

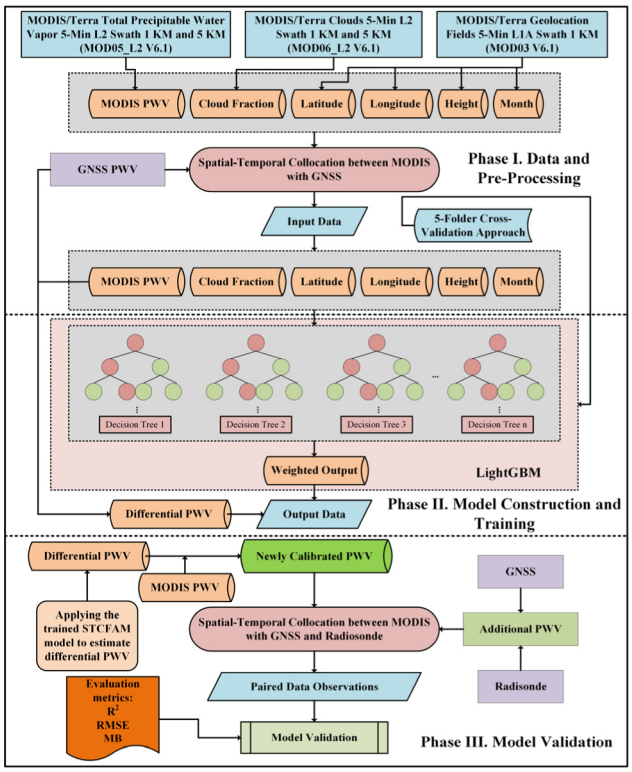


Fig. 4. Overall flowchart of the development and validation of the STCFCM based on the LightGBM machine-learning approach, which contains three major phases: I. data and preprocessing, II. model construction and training, and III. model validation.

IV. Results

The STCFCM model, developed in this research, is designed to calibrate satellite NIR water vapor observations. Once the STCFCM is trained using MODIS NIR and GNSS PWV data of 2017 at the global 5000 training stations (see Fig. 1), we then employ it to globally calibrate the operational MODIS NIR water vapor product in 2017–2019.

The validation of the calibrated PWV data is performed on a global scale using PWV from an additional 5465 GNSS testing stations in 2017–2019, as presented in Fig. 1. The collocated MODIS–GNSS data pairs in 2017–2019 at these testing stations differ from the data pairs in 2017 at the global 5000 training stations used for model training. In addition, the STCFCM-calibrated PWV estimates are also verified using 2017–2019 PWV observations from 783 radiosonde stations across the globe.

The validation process of the STCFCM is independent of its training procedure, as distinct data measurements in the spatial–temporal domain are utilized for algorithm training and validation, respectively. The evaluation results of the STCFCM are illustrated in the following subsections.

A. Comparison of MODIS NIR PWV Versus GNSS PWV

1) Overall Statistics: Fig. 5 presents the differential PWV estimates (PWV_D) in 2017–2019 across the world. The

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:33:45 UTC from IEEE Xplore. Restrictions apply.

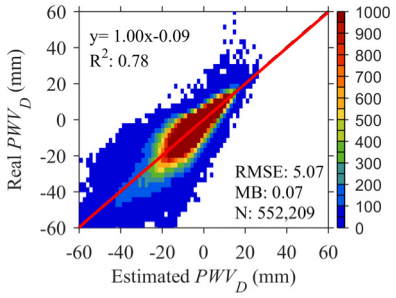


Fig. 5. Comparison of the differential water vapor data of MODIS NIR PWV from the GNSS and the STCFCM during 2017–2019 across the globe, under all weather conditions. The real PWV_D is computed using MODIS NIR PWV minus GNSS PWV. The estimated PWV_D is computed based on the STCFCM model. The color bar indicates the number of data points.

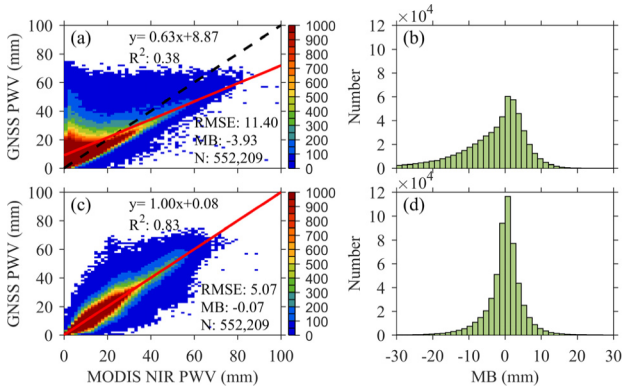


Fig. 6. Comparison of all-weather PWV data from the MOD05 and the STCFCM model, with GNSS PWV reference data during 2017–2019 across the globe. (a) Scatter plot of operational PWV against GNSS PWV, (b) histogram of MB of operational PWV against GNSS PWV, (c) scatter plot of calibrated PWV against GNSS PWV, and (d) histogram of MB of calibrated PWV against GNSS PWV. The color bar indicates the number of data points.

STCFCM-estimated differential water vapor agreed well with the real PWV_D calculated from reference GNSS-derived PWV data, showing $R^2 = 0.78$, RMSE = 5.07 mm, and MB = 0.07 mm. The linear fitting line of the comparison between real and estimated PWV_D is close to the reference 1:1 line, which implies that the STCFCM-based PWV_D is almost equal to the real GNSS-based differential water vapor data.

In Fig. 6, the MODIS-calibrated NIR water vapor observations presented a better agreement with GNSS-measured reference PWV data, compared with operational MODIS/Terra NIR water vapor product MOD05. The MOD05 all-weather PWV has poor quality, with $R^2 = 0.38$, RMSE = 11.40 mm, and MB = −3.93 mm against GNSS-retrieved water vapor observations. This evaluation result of MOD05 PWV is consistent with previous research [20]. The STCFCM-calibrated water vapor estimates presented a significant improvement: an enhancement of 118.42% in R^2 from 0.38 to 0.83, a decrease of 55.53% in RMSE from 11.40 to 5.07 mm, and a decrease of 98.22% in MB from −3.93 to −0.07 mm. The MODIS-calibrated NIR water vapor data also have a more reasonable MB distribution compared with GNSS-observed reference PWV data [see Fig. 6(b) and (d)].

As displayed in Fig. 7, the STCFCM-calibrated confident-clear PWV data perform better than the operational MOD05

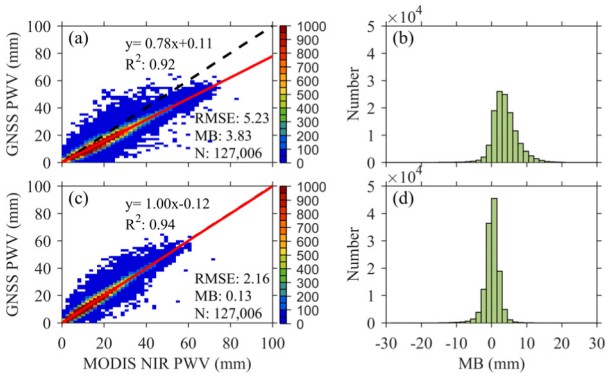


Fig. 7. Comparison of confident-clear PWV data from the MOD05 and the STCFCM model, with GNSS PWV reference data during 2017–2019 across the globe. (a) Scatter plot of operational PWV against GNSS PWV, (b) histogram of MB of operational PWV against GNSS PWV, (c) scatter plot of calibrated PWV against GNSS PWV, and (d) histogram of MB of calibrated PWV against GNSS PWV. The color bar indicates the number of data points.

water vapor product, showing an enhancement of 2.17% in R^2 from 0.92 to 0.94, a decrease of 58.70% in RMSE from 5.23 to 2.16 mm, and a decrease of 96.61% in MB from 3.83 to 0.13 mm. The STCFCM-estimated all-weather PWV data had a better water vapor measurement performance (RMSE = 5.07 mm) compared with operational MODIS/Terra NIR confident-clear water vapor measurements (RMSE = 5.23 mm). It shows that the STCFCM model, proposed in this research, can significantly improve the observational quality of worldwide PWV estimates from MODIS/Terra NIR observations, regardless of weather conditions.

2) Analysis With Cloud Fraction: The observational performance of satellite-sensed NIR PWV estimates is considerably impacted when clouds exist, as shown in Fig. 8. The slopes of liner regression lines of STCFCM-calibrated PWV are much closer to 1 under different cloud fraction conditions, implying that MODIS-calibrated NIR water vapor data are comparable to GNSS-derived reference PWV at different weather conditions.

The MODIS-corrected NIR water vapor presented an improved R^2 above 0.90 with reference GNSS PWV data when the cloud fraction was below 1, compared to operational MOD05 water vapor measurements. When the cloud fraction was equal to 1, the correlation R^2 between MODIS NIR PWV and GNSS PWV was enhanced by 162.07% from 0.29 to 0.76. The R^2 of operational MODIS/Terra NIR water vapor measurements tends to decrease with the increased cloud fraction levels, whereas the STCFCM-calibrated NIR PWV estimates show a stable correlation R^2 under different cloud fraction levels.

The STCFCM-estimated PWV results exhibit lower RMSE values at different cloud fraction levels when compared with operational MOD05 water vapor estimates. The STCFCM-calibrated NIR water vapor data had a decrease in RMSE of 45.68% under probably clear conditions, 35.19%–40.61% under probably cloudy conditions, and 32.52%–58.01% under confident cloudy conditions. Though both operational and calibrated PWV measurements tend to have a

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:33:45 UTC from IEEE Xplore. Restrictions apply.

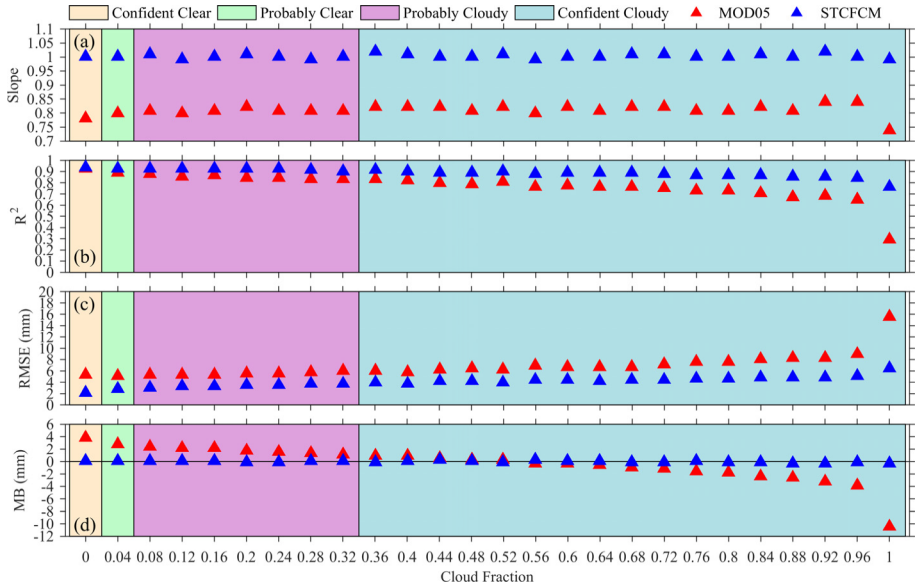


Fig. 8. Comparison of PWV data from the MOD05 and the STCFCM model, using GNSS PWV as reference data during 2017–2019 across the globe, under different cloud fraction levels. (a) Slopes of linear regression lines between MODIS PWV and GNSS PWV, (b) R^2 between MODIS PWV and GNSS PWV, (c) RMSE between MODIS PWV and GNSS PWV, and (d) MB between MODIS PWV and GNSS PWV.

larger RMSE with the increase of cloud fraction, the increase rate of calibrated PWV is, however, much smaller.

Fig. 8 shows that the MB between STCFCM-estimated PWV and GNSS PWV has reduced to approach 0 at different cloud fraction levels. The operational Terra MODIS NIR water vapor data (MOD05) products apparently overestimate the water vapor values at the lower cloud fraction levels, while in the higher fraction levels, MOD05 products underestimate PWV values.

3) Analysis With Station-Wise: Figs. 9 and 10 illustrate the station-wise comparison result between MODIS NIR PWV and GNSS PWV during 2017–2019 over the world. In Fig. 9, the operational MOD05 water vapor products have an RMSE ranging from 4 to 8 mm at most GNSS stations, except the PWV product observed under confident-cloudy and all-weather conditions.

The station-wise RMSE values of operational MOD05 water vapor data are, in general, larger than 8 mm under confident-cloudy and all-weather conditions. After the calibration using the STCFCM approach, MODIS-measured NIR water vapor estimates show smaller station-wise RMSE values at most stations under different cloud conditions. The station-wise RMSE reduction is more than 40% at most stations, regardless of cloud conditions.

The operational PWV measurements from MODIS NIR bands, derived under confident-clear, probably-clear, and probably-cloudy conditions, show positive station-wise MB values (overestimation) at most stations (see Fig. 10). On the other hand, they underestimate PWV values with negative station-wise MB under confident-cloudy and all-weather conditions. The station-wise MB of STCFCM-estimated water vapor data is approaching 0 at most stations, regardless of cloud conditions.

4) Analysis With Time-Series: The monthly time-series assessment of PWV observations from MODIS/Terra NIR channels is shown in Figs. 11 and 12. The monthly-series RMSE of STCFCM-estimated water vapor data are, in general, smaller than those of MOD05 water vapor measurements (see Fig. 11). Both operational and calibrated MODIS NIR water vapor data tend to have a higher monthly-series RMSE as cloud fraction increases, with the largest monthly-series RMSE under confident cloudy conditions. This could be because of the impact of clouds, consistent with the result shown by Xu and Liu [20]. In addition, both MOD05 and STCFCM-calibrated water vapor data present a larger monthly-series RMSE during wet months of the year, namely June, July, and August. This could be on account of larger PWV values in wet months, which can lead to larger RMSE as shown in Xu and Liu [30].

The STCFCM-calibrated water vapor estimates show a smaller monthly-series MB, as displayed in Fig. 12. The MOD05 products overestimate PWV values with positive monthly-series MB values in the months of 2017–2019 under confident-clear, probably-clear, and probably-cloudy conditions. Yet, in confident-cloudy and all-weather conditions the MOD05 PWV products have negative monthly-series MB values, underestimating water vapor values in the months of 2017–2019. The monthly-series MB of STCFCM-calibrated NIR water vapor observations is approaching 0, exhibiting a very small overestimation or underestimation of PWV values.

B. Comparison of MODIS NIR PWV Versus Radiosonde PWV

To further evaluate the performance of the STCFCM model, we also used the radiosonde-measured reference PWV data during 2017–2019 across the world to evaluate the accuracy of PWV products from MODIS NIR observations, before and after the employment of the STCFCM.

As presented in Fig. 13, the STCFCM-calibrated all-weather PWV estimates correlated 0.87, an RMSE of 5.76 mm, and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:33:45 UTC from IEEE Xplore. Restrictions apply.

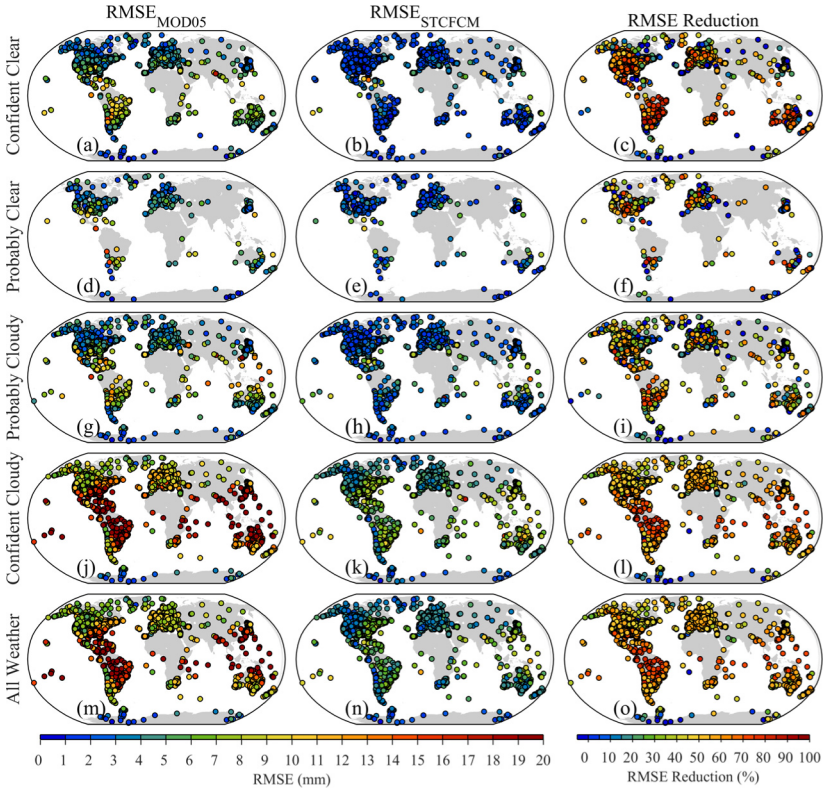


Fig. 9. Station-wise RMSE comparison of PWV data from MOD05 and the STCFCM, using GNSS PWV as reference data during 2017–2019 across the globe, under (a)–(c) confident clear, (d)–(f) probably clear, (g)–(i) probably cloudy, (j)–(l) confident cloudy, and (m)–(o) all weather conditions. The first, second, and third columns are RMSE of operational PWV, RMSE of calibrated PWV, and RMSE reduction of calibrated PWV versus operational PWV, respectively.

an MB of 0.12 mm compared to radiosonde-derived reference PWV, which are superior to operational MOD05 PWV products ($R^2 = 0.38$, RMSE = 14.67 mm, and MB = −5.86 mm). The RMSE between MODIS PWV and radiosonde PWV has reduced by 60.74% from 14.67 to 5.76 mm. This RMSE reduction is larger than the reduction in RMSE between MODIS PWV with GNSS PWV, that is, 55.53% from 11.40 to 5.07 mm.

At different cloud fraction levels, the RMSE values of operational MOD05 PWV observations are in the range of 6.02–19.40 mm. The newly calibrated PWV data show an RMSE of 3.47–6.69 mm. After the employment of the STCFCM, the RMSE of MODIS PWV measurements has been decreased at each cloud fraction condition.

Temporally, the newly calibrated PWV estimates also show smaller RMSE values in all months of 2017–2019, when compared with operational PWV observations, implying the stability of the STCFCM model. At most radiosonde stations, the STCFCM-based PWV estimates have lower station-wise RMSE values than operational MOD05 PWV products.

V. Discussion

A novel STCFCM method has been developed to calibrate the PWV estimates from satellite NIR bands for the first time. The widely used MODIS NIR measurements are used in this work to test the performance of the STCFCM.

In our study, the STCFCM model is trained based on GNSS PWV data of 2017 from global 5000 training stations (see Fig. 1) using a fivefold cross-validation approach. Once it is well trained, we then apply it to globally calibrate the operational MODIS NIR water vapor products in 2017–2019.

The validation of the newly calibrated PWV data is conducted on a global scale using PWV from independent 5465 GNSS testing stations and 783 radiosonde testing stations, both collected in 2017–2019 (see Fig. 1). The collocated MODIS–GNSS or MODIS–radiosonde data pairs used for evaluation are independent of the data pairs in 2017 at the global 5000 GNSS training stations used for model training.

It should be noted that several GNSS and radiosonde stations, which do not satisfy the spatial–temporal matching criteria between MODIS and GNSS (radiosonde), are not selected and used in our model training and validation procedure. Therefore, the number of GNSS (radiosonde) stations used in our spatial station-wise verification results, shown in Fig. 13, is smaller than that in Fig. 1.

A. Spatial and Temporal Stabilities of the STCFCM

The validation data at testing stations is spatially different from the training data at training stations. The validation data for 2018–2019 is temporally different from the training data for 2017. As a result, the performance of the STCFCM developed in this article is verified independently in the spatial–temporal domain.

Spatially, the station-wise RMSE and MB discrepancies between MOD05 PWV with GNSS/radiosonde PWV are reduced at most stations across the world, as presented in

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:33:45 UTC from IEEE Xplore. Restrictions apply.

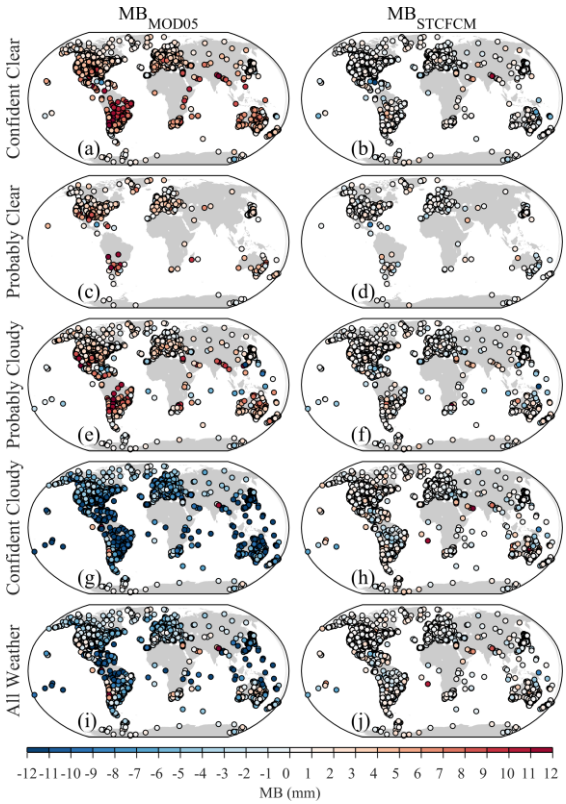


Fig. 10. Station-wise MB comparison of PWV data from the MOD05 and the STCFCM, using GNSS PWV as reference data during 2017–2019 across the globe, under (a) and (b) confident clear, (c) and (d) probably clear, (e) and (f) probably cloudy, (g) and (h) confident cloudy, and (i) and (j) all weather conditions. The first and second columns are MB of operational PWV and MB of calibrated PWV, respectively.

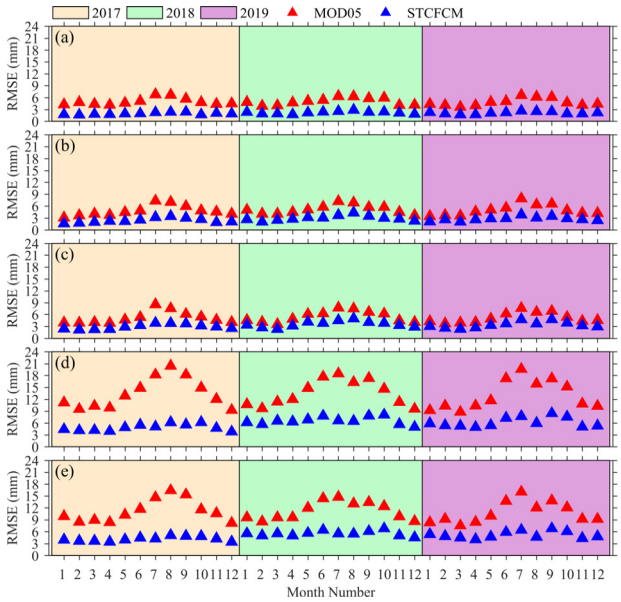


Fig. 11. Monthly-series RMSE comparison of PWV data from the MOD05 and the STCFCM model, using GNSS PWV as reference data during 2017–2019 across the globe, under (a) confident clear, (b) probably clear, (c) probably cloudy, (d) confident cloudy, and (e) all weather conditions.

Figs. 9 and 13. This illustrates that the STCFCM, developed in this research, can considerably improve the performance of operational PWV satellite products under all-weather conditions in most areas of the globe. This could be because the

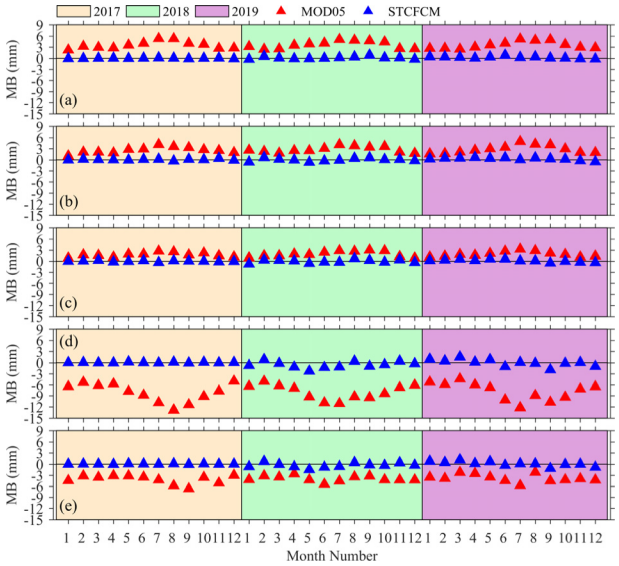


Fig. 12. Monthly-series MB comparison of PWV data from the MOD05 and the STCFCM model, using GNSS PWV as reference data during 2017–2019 across the globe, under (a) confident clear, (b) probably clear, (c) probably cloudy, (d) confident cloudy, and (e) all weather conditions.

STCFCM has employed the spatial field data (i.e., latitude, longitude, and height).

In addition, both operational and calibrated PWV estimates have larger RMSE values at coastal areas globally, due to the magnitude of PWV values. This result shows a good consistency with those in previous studies [47], [48]. It should be mentioned that the STCFCM also presents a slightly smaller RMSE value at regions with more training stations, which is consistent with the result in Xu and Liu [48].

In the temporal domain, the STCFCM-calibrated PWV estimates exhibit smaller RMSE values in the months of 2017–2019, as shown in Figs. 11 and 13. In 2017, the newly calibrated all-weather PWV data show a monthly RMSE in the range from 3.43 mm (December) to 5.08 mm (August) compared to GNSS PWV and from 4.80 mm (March) to 6.42 mm (August) compared to radiosonde PWV. In the other two independent years, that is, 2018–2019, the monthly RMSE values of STCFCM-estimated all-weather PWV data are in the range between 4.04 mm (April 2019) and 6.79 mm (September 2019) and between 4.78 mm (March 2019) and 6.63 mm (September 2018) compared with GNSS and radiosonde PWV, respectively. In terms of RMSE, the performance of STCFCM in 2018–2019 is comparable to that in 2017 (i.e., same as the training period). This suggests that the STCFCM, developed in this article, shows very good temporal stability periods, which could apply to other time periods.

Notedly, both MOD05 and STCFCM-calibrated PWV products have higher monthly RMSE values during wet months. This could be on account of the large PWV values in wet months [30].

B. Comparison Between the STCFCM With Previous Algorithms

The STCFCM-calibrated all-weather water vapor product shows $R^2 = 0.83$, RMSE = 5.07 mm, and MB = −0.07 mm

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:33:45 UTC from IEEE Xplore. Restrictions apply.

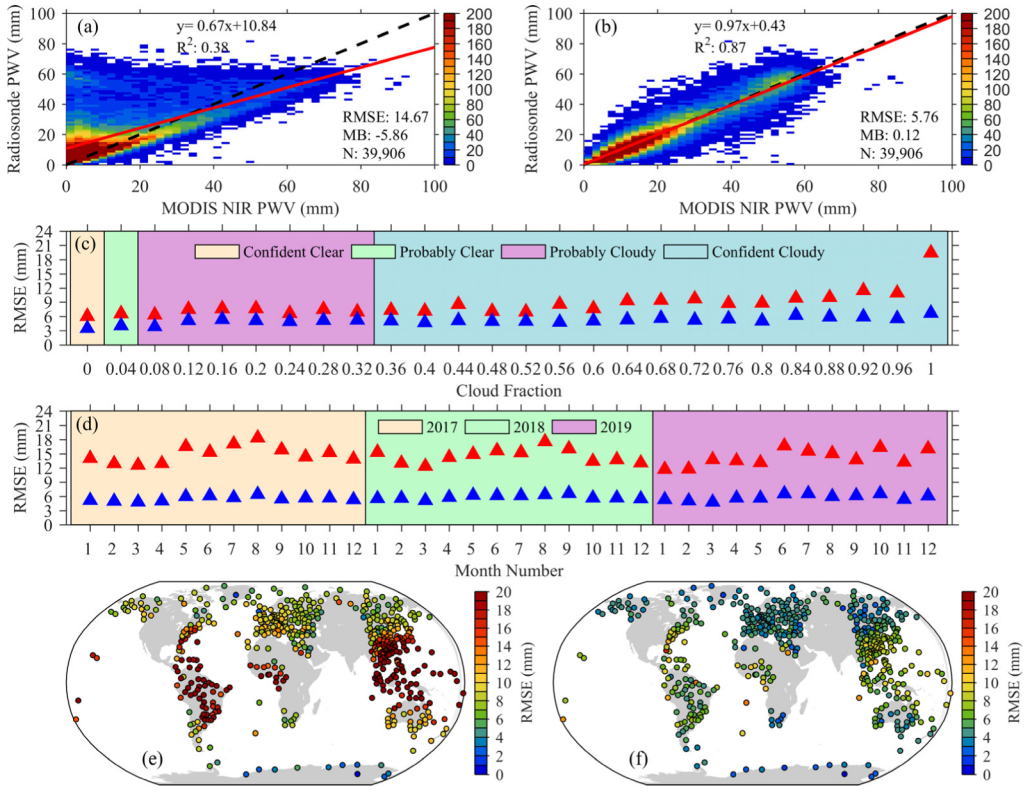


Fig. 13. Comparison of all-weather PWV data from the MOD05 and the STCFCM model, using radiosonde PWV as a reference during 2017–2019 across the globe. (a) Scatter plot of operational all-weather PWV against radiosonde PWV, (b) scatter plot of calibrated all-weather PWV against radiosonde PWV, (c) RMSE of operational/calibrated PWV against radiosonde PWV under different cloud fraction levels and different cloud conditions, (d) monthly-series RMSE of operational/calibrated all-weather PWV against radiosonde PWV, (e) station-wise RMSE of operational all-weather PWV against radiosonde PWV, and (f) station-wise RMSE of calibrated all-weather PWV against radiosonde PWV. The color bar in (a) and (b) indicates the number of data points. In (c) and (d), the red triangle indicates the operational MOD05 PWV, while the blue triangle indicates the calibrated PWV.

compared to GNSS-measured reference PWV observations. The newly calibrated PWV estimates also have a better consistency with radiosonde-observed reference PWV data, with R^2, RMSE, and MB of 0.87, 5.76 mm, and 0.12 mm, respectively. In addition, the calibrated PWV data, observed under all weather conditions, perform better (i.e., smaller RMSE) than operational MODIS/Terra NIR confident-clear PWV products, as displayed in Figs. 6 and 7. This implies that the STCFCM is capable and effective in calibrating all-weather PWV satellite products.

The STCFCM model is developed and validated across the globe, which is different from the previous research conducted in a regional area [24], [32], [33], [49], [50]. Additionally, the methods, developed in previous studies, are trained and validated using the same GNSS stations [24], [32], [33], [49], [51], while our STCFCM method is trained and validated based on completely different GNSS and radiosonde sites. Regardless of the above differences between our research with previous research, the STCFCM model developed in this article still exhibits a comparable or better performance on operational MODIS NIR water vapor estimates compared with previous studies [24], [33], [49].

C. Current Problem and Future Research

While the newly proposed STCFCM method can improve the observational quality of operational PWV satellite products at each cloud fraction level, it also presents an increased RMSE value with the increment of cloud fraction, as the result displayed in Figs. 8 and 13. This implies that the calibration performance of the STCFCM is still affected by the presence of clouds.

The STCFCM approach is developed using global GNSS data of 2017 over land areas. The verification of the STCFCM is also primarily conducted based on GNSS and radiosonde data over land regions globally. This is because very few GNSS and radiosonde stations are available over oceanic areas globally, as presented in Fig. 1. Additionally, the performance of the STCFCM is validated using limited GNSS and radiosonde stations in some particular regions, such as Antarctica, China, Russia, and so forth.

In future research, we will further reduce the influence of clouds on the STCFCM by incorporating additional data sources. At the same time, we will continue to refine the STCFCM over oceanic areas using high-quality ocean PWV data. We will also continue to evaluate the performance of the STCFCM in the regions with limited GNSS and radiosonde stations. Moreover, we will further verify the performance of the STCFCM on other satellite-retrieved NIR PWV products, in addition to the MODIS sensor.

VI. Conclusion

We develop a novel STCFCM approach for operational satellite-sensed NIR PWV data to generate calibrated PWV estimates. The performance of the STCFCM is evaluated based

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:33:45 UTC from IEEE Xplore. Restrictions apply.

on reference PWV from GNSS and radiosonde, independent of the GNSS-based training data in both spatial and temporal dimensions. Major findings are summarized as follows.

1) The STCFCM-calibrated all-weather PWV estimates perform better than operational PWV products. The RMSE is reduced by 55.53% from 11.40 to 5.07 mm and 60.74% from 14.67 to 5.76 mm compared to GNSS and radiosonde PWV, respectively.
2) The newly calibrated PWV data present smaller RMSE values at each different cloud fraction level. When the cloud fraction is below 0.01, that is, confident clear, the RMSE is reduced by 58.70% from 5.23 to 2.16 mm compared with GNSS-based PWV data.
3) The STCFCM-estimated PWV data, observed under all weather conditions, outperform operational MODIS confident-clear PWV products, denoting the capability and effectiveness of the STCFCM in calibrating all-weather PWV satellite products.
4) The performance of the STCFCM is overall stable in the spatial–temporal domain, which could be because both spatial and temporal fields are utilized in the STCFCM. The newly proposed STCFCM method has great potential to apply to other temporal periods as well as other satellite NIR water vapor products.

ACKNOWLEDGMENT

The authors gratefully thank the two anonymous reviewers and the editorial team for their invaluable comments that helped improve this manuscript a lot. They would like to thank the National Aeronautics and Space Administration (NASA) Level-1 and Atmosphere Archive and Distribution System (LAADS) Distributed Active Archive Center (DAAC) for providing the MODIS data products. They would also like to thank NGL for providing the GNSS water vapor data. The radiosonde observations, obtained from the Integrated Global Radiosonde Archive Version 2, are also acknowledged.

REFERENCES

[1] I. M. Held and B. J. Soden, "Water vapor feedback and global warming," *Annu. Rev. Energy Environ.*, vol. 25, no. 1, pp. 441–475, Nov. 2000.

[2] B. J. Soden, R. T. Wetherald, G. L. Stenchikov, and A. Robock, "Global cooling after the eruption of mount pinatubo: A test of climate feedback by water vapor," *Science*, vol. 296, no. 5568, pp. 727–730, Apr. 2002.

[3] T. Schneider, P. A. O'Gorman, and X. J. Levine, "Water vapor and the dynamics of climate changes," *Rev. Geophys.*, vol. 48, no. 3, Jul. 2010.

[4] L. Bengtsson, K. I. Hodges, S. Koumoutsaris, M. Zahn, and N. Keenlyside, "The changing atmospheric water cycle in polar regions in a warmer climate," *Tellus A, Dyn. Meteorol. Oceanogr.*, vol. 63, no. 5, pp. 907–920, Jan. 2011.

[5] W. P. Elliott and D. J. Gaffen, "On the utility of radiosonde humidity archives for climate studies," *Bull. Amer. Meteorol. Soc.*, vol. 72, no. 10, pp. 1507–1520, Oct. 1991.

[6] K. Gui et al., "Water vapor variation and the effect of aerosols in China," *Atmos. Environ.*, vol. 165, pp. 322–335, Sep. 2017, doi: 10.1016/j.atmosenv.2017.07.005.

[7] B. Chen and Z. Liu, "Global water vapor variability and trend from the latest 36 year (1979 to 2014) data of ECMWF and NCEP reanalyses, radiosonde, GPS, and microwave satellite," *J. Geophys. Res., Atmos.*, vol. 121, no. 19, p. 11, Oct. 2016, doi: 10.1002/2016jd024917.

[8] K. E. Trenberth, J. Fasullo, and L. Smith, "Trends and variability in column-integrated atmospheric water vapor," *Climate Dyn.*, vol. 24, nos. 7–8, pp. 741–758, Jun. 2005.

[9] W. Rohm, J. Guzikowski, K. Wilgan, and M. Kryza, "4DVAR assimilation of GNSS zenith path delays and precipitable water into a numerical weather prediction model WRF," *Atmos. Meas. Techn.*, vol. 12, no. 1, pp. 345–361, Jan. 2019.

[10] R. Saunders, "Assimilation of OLCI total column water vapour in the met office global numerical weather prediction system," *Meteorol. Appl.*, vol. 28, no. 5, p. e2029, Sep. 2021.

[11] H. Hersbach, B. Bell, P. Berrisford, S. Hirahara, and J. Thépaut, "The ERA5 global reanalysis," *Quart. J. Roy. Meteorol. Soc.*, vol. 146, no. 730, pp. 1999–2049, 2020.

[12] S. Manandhar, S. Dev, Y. H. Lee, Y. S. Meng, and S. Winkler, "A data-driven approach for accurate rainfall prediction," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 9323–9331, Nov. 2019.

[13] B. N. Holben et al., "AERONET—A federated instrument network and data archive for aerosol characterization," *Remote Sens. Environ.*, vol. 66, no. 1, pp. 1–16, Oct. 1998.

[14] H. Hu, R. Yang, W.-C. Lee, Y. Cao, J. Mao, and L. Gao, "Multi-sensor study of precipitable water vapor and atmospheric profiling from microwave radiometer, GNSS/MET, radiosonde, and ECMWF reanalysis in Beijing," *J. Appl. Remote Sens.*, vol. 14, no. 4, Dec. 2020, Art. no. 044514.

[15] B. J. Soden, D. D. Turner, B. M. Lesht, and L. M. Miloshevich, "An analysis of satellite, radiosonde, and LiDAR observations of upper tropospheric water vapor from the atmospheric radiation measurement program," *J. Geophys. Res., Atmos.*, vol. 109, no. D4, Feb. 2004, Art. no. D04105.

[16] J. Vaquero-Martínez et al., "Inter-comparison of integrated water vapor from satellite instruments using reference GPS data at the Iberian peninsula," *Remote Sens. Environ.*, vol. 204, pp. 729–740, Jan. 2018.

[17] J. Xu and Z. Liu, "The first validation of Sentinel-3 OLCI integrated water vapor products using reference GPS data in Mainland China," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4102817.

[18] J. Vaquero-Martínez, M. Antón, J. P. O. de Galisteo, R. Román, V. E. Cachorro, and D. Mateos, "Comparison of integrated water vapor from GNSS and radiosounding at four GRUAN stations," *Sci. Total Environ.*, vol. 648, pp. 1639–1648, Jan. 2019.

[19] J. Wang and L. Zhang, "Systematic errors in global radiosonde precipitable water data from comparisons with ground-based GPS measurements," *J. Climate*, vol. 21, no. 10, pp. 2218–2238, May 2008.

[20] J. Xu and Z. Liu, "Evaluation of precipitable water vapor product from MODIS and MERSI-II NIR channels using Ground- based GPS measurements over Australia," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 8744–8758, 2022.

[21] C. Yu, Z. Li, and G. Blewitt, "Global comparisons of ERA5 and the operational HRES tropospheric delay and water vapor products with GPS and MODIS," *Earth Space Sci.*, vol. 8, no. 5, May 2021, Art. no. e2020EA001417.

[22] X. Ma, Y. Yao, B. Zhang, Y. Qin, Q. Zhang, and H. Zhu, "An improved MODIS NIR PWV retrieval algorithm based on an artificial neural network considering the land-cover types," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5622412.

[23] J. Xu and Z. Liu, "Radiance-based retrieval of total water vapor content from Sentinel-3A OLCI NIR channels using ground-based GPS measurements," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 104, Dec. 2021, Art. no. 102586.

[24] D. Zhu, K. Zhang, L. Yang, S. Wu, and L. Li, "Evaluation and calibration of MODIS near-infrared precipitable water vapor over China using GNSS observations and ERA-5 reanalysis dataset," *Remote Sens.*, vol. 13, no. 14, p. 2761, Jul. 2021.

[25] G. Di Natale et al., "Implementation and validation of a retrieval algorithm for profiling of water vapor from differential attenuation measurements at microwaves," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5939–5948, Aug. 2019.

[26] S. W. Seemann, J. Li, W. P. Menzel, and L. E. Gumley, "Operational retrieval of atmospheric temperature, moisture, and ozone from MODIS infrared radiances," *J. Appl. Meteorol.*, vol. 42, no. 8, pp. 1072–1091, Aug. 2003.

[27] H. Wang, A. H. Souri, G. González Abad, X. Liu, and K. Chance, "Ozone monitoring instrument (OMI) total column water vapor version 4 validation and applications," *Atmos. Meas. Techn.*, vol. 12, no. 9, pp. 5183–5199, Sep. 2019.

[28] J. Xu and Z. Liu, "A back propagation neural network-based algorithm for retrieving all-weather precipitable water vapor from MODIS NIR measurements," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5633614.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:33:45 UTC from IEEE Xplore. Restrictions apply.

[29] M. D. King, Y. J. Kaufman, W. P. Menzel, and D. Tanre, "Remote sensing of cloud, aerosol, and water vapor properties from the moderate resolution imaging spectrometer (MODIS)," *IEEE Trans. Geosci. Remote Sens.*, vol. 30, no. 1, pp. 2–27, Jan. 1992.

[30] J. Xu and Z. Liu, "Water vapour products from ERA5, MERSI-II/FY-3D, OLCI/Sentinel-3A, OLCI/Sentinel-3B, MODIS/Aqua and MODIS/Terra in Australia: A comparison against in situ GPS water vapour data," *Quart. J. Roy. Meteorol. Soc.*, vol. 149, no. 753, pp. 1435–1458, Apr. 2023.

[31] J. Xu and Z. Liu, "A back propagation neural network-based calibration approach for Sentinel-3 OLCI near-infrared water vapor product," *IEEE Geosci. Remote Sens. Lett.*, vol. 20, pp. 1–5, 2023.

[32] J. Xu and Z. Liu, "A linear regression of differential PWV calibration model to improve the accuracy of MODIS NIR all-weather PWV products based on ground-based GPS PWV data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 7929–7951, 2022.

[33] J. Xu and Z. Liu, "Improving the accuracy of MODIS near-infrared water vapor product under all weather conditions based on machine learning considering multiple dependence parameters," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4101115.

[34] Y. J. Kaufman and B.-C. Gao, "Remote sensing of water vapor in the near IR from EOS/MODIS," *IEEE Trans. Geosci. Remote Sens.*, vol. 30, no. 5, pp. 871–884, Sep. 1992.

[35] V. V. Salomonson, W. L. Barnes, P. W. Maymon, H. E. Montgomery, and H. Ostrow, "MODIS: Advanced facility instrument for studies of the Earth as a system," *IEEE Trans. Geosci. Remote Sens.*, vol. 27, no. 2, pp. 145–153, Mar. 1989.

[36] B. Gao and Y. J. Kaufman, "Water vapor retrievals using Moderate Resolution Imaging Spectroradiometer (MODIS) near-infrared channels," *J. Geophys. Res., Atmos.*, vol. 108, no. D13, Jul. 2003.

[37] S. Platnick et al., "The MODIS cloud products: Algorithms and examples from Terra," *IEEE Trans. Geosci. Remote Sens.*, vol. 41, no. 2, pp. 459–473, Feb. 2003, doi: 10.1109/TGRS.2002.808301.

[38] G. Blewitt, W. Hammond, and C. Kreemer, "Harnessing the GPS data explosion for interdisciplinary science," *Eos*, vol. 99, p. 485, Sep. 2018.

[39] W. Bertiger et al., "GipsyX/RTGx, a new tool set for space geodetic operations and research," *Adv. Space Res.*, vol. 66, no. 3, pp. 469–489, Aug. 2020.

[40] M. Bevis, S. Businger, T. A. Herring, C. Rocken, R. A. Anthes, and R. H. Ware, "GPS meteorology: Remote sensing of atmospheric water vapor using the global positioning system," *J. Geophys. Res.*, vol. 97, no. D14, pp. 15787–15801, Oct. 1992.

[41] M. Bevis et al., "GPS meteorology: Mapping zenith wet delays onto precipitable water," *J. Appl. Meteorol.*, vol. 33, no. 3, pp. 379–386, Mar. 1994.

[42] I. Durre, X. Yin, R. S. Vose, S. Applequist, and J. Arnfield, "Enhancing the data coverage in the integrated global radiosonde archive," *J. Atmos. Ocean. Technol.*, vol. 35, no. 9, pp. 1753–1770, Sep. 2018.

[43] Y. Zhang, C. Cai, B. Chen, and W. Dai, "Consistency evaluation of precipitable water vapor derived from ERA5, ERA-interim, GNSS, and radiosondes over China," *Radio Sci.*, vol. 54, no. 7, pp. 561–571, Jul. 2019.

[44] M. Antón, D. Loyola, R. Román, and H. Vömel, "Validation of GOME-2/MetOp-A total water vapour column using reference radiosonde data from the GRUAN network," *Atmos. Meas. Techn.*, vol. 8, no. 3, pp. 1135–1145, Mar. 2015.

[45] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, La Jolla, CA, USA, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds., 2017. Accessed: May 14, 2022. [Online]. Available: https://www.webofscience.com/wos/woscc/full-record/WOS:000452649403021

[46] J. H. Friedman, "Stochastic gradient boosting," *Comput. Statist. Data Anal.*, vol. 38, no. 4, pp. 367–378, Feb. 2002.

[47] J. Xu and Z. Liu, "A gradient boosting decision tree based correction model for AIRS infrared water vapor product," *Geophys. Res. Lett.*, vol. 50, no. 14, Jul. 2023, Art. no. e2023GL104072.

[48] J. Xu and Z. Liu, "Enhanced all-weather precipitable water vapor retrieval from MODIS near-infrared bands using machine learning," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 114, Nov. 2022, Art. no. 103050.

[49] Z. Xiong, X. Sun, J. Sang, and X. Wei, "Modify the accuracy of MODIS PWV in China: A performance comparison using random forest, generalized regression neural network and back-propagation neural network," *Remote Sens.*, vol. 13, no. 11, p. 2215, Jun. 2021.

[50] J. Xu and Z. Liu, "Long-term calibration of satellite-based all-weather precipitable water vapor product from FengYun-3A MERSI near-infrared bands from 2010 to 2017 in China," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023.

[51] J. Xu, Z. Liu, G. Hong, and Y. Cao, "A new machine-learning-based calibration scheme for MODIS thermal infrared water vapor product using BPNN, GBDT, GRNN, KNN, MLPNN, RF, and XGBoost," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5001412, doi: 10.1109/TGRS.2024.3356578.



Jiafei Xu received the B.Sc. degree in remote-sensing science and technology from Shandong Agricultural University, Tai'an, China, in 2015, and the M.Sc. degree in geomatics engineering from the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China, in 2019. He is currently pursuing the Ph.D. degree with the Department of Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University, Hong Kong.

His research interests include the algorithm development for enhancing precipitable water vapor retrievals from multisatellite visible, near-infrared, infrared, and microwave data observations as well as the algorithm validation for various ground-based, satellite-based, and reanalysis-based precipitable water vapor data products. As of present, he has done several works associated with satellite water vapor observations, and they have been published in several peer-reviewed journals, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, *International Journal of Applied Earth Observation and Geoinformation*, *Quarterly Journal of the Royal Meteorological Society*, and *Geophysical Research Letters*.



Zhizhao Liu (Member, IEEE) received the B.Sc. degree in surveying engineering from the Jiangxi University of Science and Technology, Ganzhou, China, in 1994, the M.Sc. degree in geodesy from Wuhan University, Wuhan, China, in 1997, and the Ph.D. degree in geomatics engineering from the University of Calgary, Calgary, AB, Canada, in 2004.

He is currently a Professor with the Department of Land Surveying and Geo-Informatics, The Hong Kong Polytechnic University, Hong Kong. His research interests include new algorithm development for precise global positioning system (GPS) and global navigation satellite system (GNSS), GPS/GNSS precise point positioning (PPP), ionosphere modeling and scintillation monitoring, tropospheric remote sensing and modeling, and GPS/GNSS meteorology. He has over 20 years of experience in GPS/GNSS research. His group has developed a highly efficient and effective algorithm for cycle slip detection and repair for dual- and multifrequency GNSS carrier phase data. The algorithm his group developed can improve the accuracy of water vapor retrieval from remote-sensing satellite data by up to 50%. His group developed China's first GPS PPP-based precipitable water vapor real-time monitoring system in the Pearl River Delta region in 2012. In 2012, his group established Hong Kong's first GPS/GNSS-based ionosphere scintillation monitoring system (two stations deployed in South and North Hong Kong) with his collaborators. His research group established Hong Kong's first GPS/GNSS-radiosonde water vapor-sounding collocation system in 2013 in collaboration with the Hong Kong Observatory.

Dr. Liu was a recipient of the Inaugural Early Career Award of the Hong Kong Research Grants Council (RGC), Hong Kong, in 2012, and the Inaugural Best Conference Paper of the China Satellite Navigation Conference (CSNC), China, in 2013. In 2014, he was nominated by the Hong Kong Observatory for the World Meteorological Organization (WMO) "Norbert Gerbier-MUMM International Award for 2015" for his paper that developed a method to evaluate the absolute accuracy of water vapor measurements.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 16:33:45 UTC from IEEE Xplore. Restrictions apply.